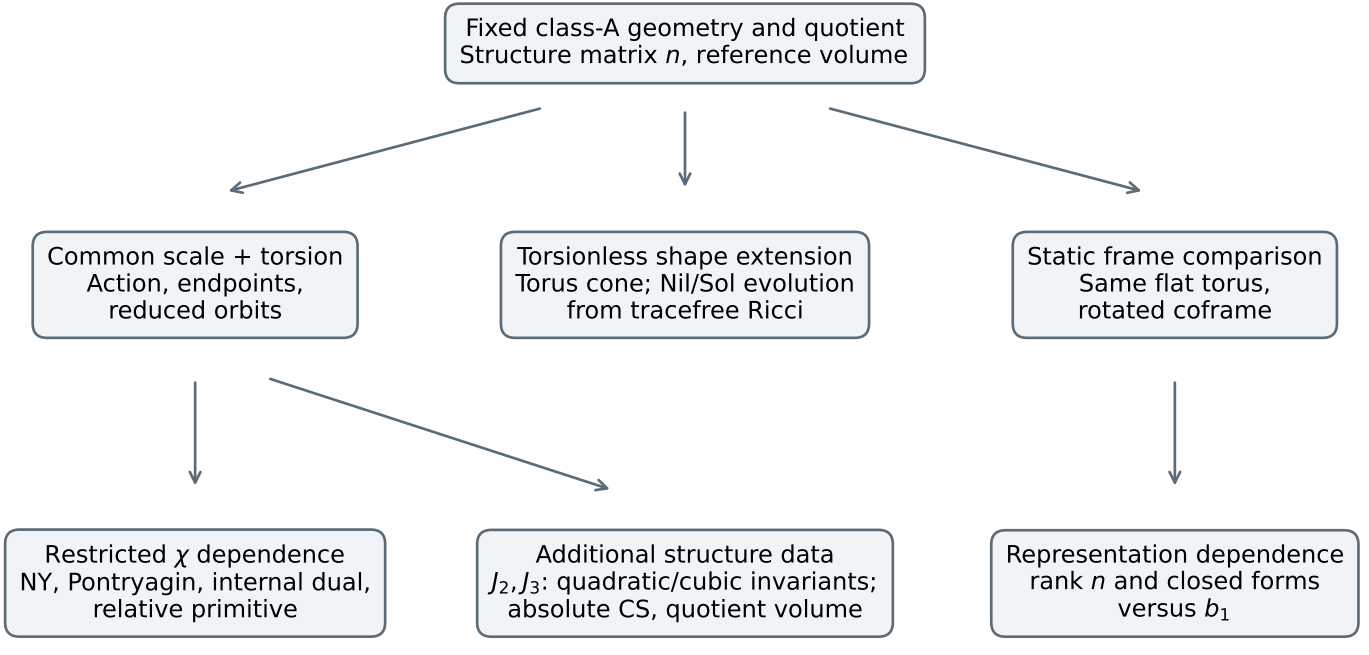
Fixed class-A geometry and quotient
Structure matrix n, reference volume
Common scale + torsion
Action, endpoints,
reduced orbits
Torsionless shape extension
Torus cone; Nil/Sol evolution
from tracefree Ricci
Static frame comparison
Same flat torus,
rotated coframe
Restricted χ dependence
NY, Pontryagin, internal dual,
relative primitive
Additional structure data
J_2,J_3: quadratic/cubic invariants;
absolute CS, quotient volume
Representation dependence
rank n and closed forms
versus b_1